

Chapter 81 — Minimally Invasive Retroperitoneal Lymph Node Dissection

- **Retroperitoneal lymph node dissection (RPLND)** is a central component of treatment for **testicular germ cell tumours (TGCTs)**.
- **Treatment goals in early-stage TGCT** (stage I and low-volume stage II): **maintain excellent oncological outcomes**, **minimise dual therapy** (surgery plus chemotherapy), and **minimise long-term treatment-related harm**.
- **Guideline-directed indications, chemotherapy-naïve**: serum tumour marker-negative patients with **stage I nonseminomatous germ cell tumour (NSGCT)**, and **small-volume stage II seminoma and NSGCT**.
- **RPLND is both diagnostic and therapeutic** in these settings:
 - **Accurate staging and cure of occult metastases** in clinical stage I disease.
 - **Cure of modest-volume retroperitoneal metastases** in clinical stage II disease.
 - **Avoidance of systemic therapy** in the 20–30% of patients with enlarged retroperitoneal nodes who prove **pN0** at RPLND.
- **Postchemotherapy indications** (after appropriate risk-directed chemotherapy): **NSGCT with residual masses >1 cm**, and **select seminomatous germ cell tumours (GCTs)**.
- **Why toxicity reduction drives the field**: once platinum-based chemotherapy made metastatic GCT curable, awareness of its **long-term toxicities** grew, prompting efforts to reduce treatment burden and protect quality of life:
 - **Fewer chemotherapy cycles, or omission of bleomycin**, in **International Germ Cell Cancer Collaborative Group (IGCCCG) good-risk** disease.
 - **Elimination of routine suprahilar dissection** from RPLND.
 - **Nerve-sparing technique** to preserve ejaculatory function.
 - **Templates** to reduce the morbidity of RPLND.
 - **Minimally invasive RPLND (MI-RPLND)** — the newest of these endeavours.
- **Purely laparoscopic RPLND is technically demanding** — reserved for centres with high-volume laparoscopic surgeons and high GCT volume. **Robotic surgery reduced some of the technical demand**, and robotic experience is growing rapidly.

KEY POINTS

- MI-RPLND aims to **minimise treatment-related toxicity** and its impact on quality of life in GCT.
- Purely laparoscopic RPLND is a **technically demanding** operation.
- **Dissemination of robotic platforms** has renewed interest in MI-RPLND.

Concept — *Because TGCT patients are young and almost always cured, the field is judged on **cure preserved at the lowest lifetime toxicity**; MI-RPLND is one more de-escalation measure and must be weighed against that standard, not against convenience.*

RETROPERITONEAL LYMPH NODE DISSECTION: AN ENDURING OPERATION WITH AN EVOLVING APPROACH

Open era

- **Classic access:** midline xiphoid-to-pubis incision; thoracoabdominal incision for suprahilar or retrocrural disease. Excellent exposure, but **prolonged convalescence**.
- **Historical perioperative care compounded morbidity:** full bowel preparation and prolonged fasting → excessive intravenous fluids and non-contemporary anaesthetics → routine nasogastric tubes, opioid-only analgesia, limited mobilisation and neglected chest physiotherapy.
- Morbidity was **accepted as the price of excellent long-term outcomes** in young survivors; **adequate exposure remains critical**.

Approach-agnostic advances

- **Prehabilitation, optimised anaesthetic protocols, extended-release non-opioid analgesia and Enhanced Recovery After Surgery (ERAS) protocols** improved convalescence after RPLND, as after radical cystectomy, prostatectomy and radical nephrectomy with inferior vena cava (IVC) thrombectomy.
- **Minimally invasive surgery (MIS)** was attractive for its reported **lower blood loss, better convalescence, better cosmesis and faster return to normal activity**.

Laparoscopic era

- The **multiquadrant, technically demanding** nature of RPLND delayed laparoscopic adoption; expert laparoscopic GCT centres pursued it nonetheless.
- **First laparoscopic RPLND: 1992** — an extended modified template case report, with minimal blood loss and a disease-free state at early follow-up.
- **Subsequent series versus open RPLND: shorter hospital stay and less blood loss, but longer operative time;** acceptable node yield, short- and intermediate-term oncological outcomes and complication profiles.
- **Why the evidence stayed weak:**
 - **Rarity of the indication** and few centres willing to operate laparoscopically.
 - **Early series were mostly stage I patients**, many of whom may have been **cured by orchidectomy alone**.
 - **Steep learning curve** limited adoption.
 - **Adjuvant chemotherapy after limited nodal involvement** confounded judgement of oncological efficacy.

Robotic era

- Robotic platforms spread rapidly through urology and training.

LAPAROSCOPIC VERSUS ROBOTIC RPLND — WHAT THE ROBOT ADDS

Feature	Pure laparoscopy	Robotic
Visualisation	Standard	Enhanced
Retraction	Assistant-dependent	Reliable fourth-arm retraction
Instrumentation	Straight instruments	Wristed instruments
Suturing and bleeding control	Difficult	Easier suturing; potentially better management of haemorrhage

Feature	Pure laparoscopy	Robotic
Circumferential great-vessel dissection	Limited	Feasible — lumbar vessels controlled
Ergonomics	Poorer	Better

- **First robotic RPLND: 2006** — short operative time, low blood loss, **2-day length of stay**.
- Case series and multi-institutional collaborations have since examined **feasibility, safety and oncological efficacy** in both **chemotherapy-naïve and postchemotherapy** settings (Table 81.1), and the approach now appears in international guidelines.

Concept — RPLND morbidity has been reduced along **two independent axes** — perioperative care (ERAS, analgesia, prehabilitation) and access (open → laparoscopic → robotic); the first benefits every approach, so the **incremental gain attributable to access alone** is smaller than it appears.

MINIMALLY INVASIVE RETROPERITONEAL LYMPH NODE DISSECTION IN NATIONAL AND INTERNATIONAL GUIDELINES

GUIDELINE POSITIONS ON MI-RPLND

Guideline body	Position on MI-RPLND	Condition attached
National Comprehensive Cancer Network (NCCN)	Silent — no comment for or against	—
American Urological Association (AUA), early-stage GCT	May be offered	Surgeon with GCT experience and MIS expertise ; patient told of the lack of long-term oncological data
European Association of Urology (EAU)	Consider only in specialist centres	Specialist testicular cancer centre with expertise in both open RPLND and MIS

- **Two of three** major guidelines address MI-RPLND; **none endorses it unconditionally**.

CONTROVERSY SURROUNDING MINIMALLY INVASIVE RETROPERITONEAL LYMPH NODE DISSECTION

- For radical and partial nephrectomy, radical cystectomy and radical prostatectomy, **high-quality evidence shows MIS is noninferior** to open surgery with better convalescence. **No such evidence exists for MI-RPLND**, owing to technical difficulty, disease rarity, an uncommon indication, and concerns raised by high-volume centres of excellence.
- MI-RPLND is therefore **the most controversial operation in urological oncology**.

THE CASE AGAINST MI-RPLND — PRINCIPAL CONCERNS

Concern	Basis	Consequence
1. Inadequate, staging-only dissection	Early laparoscopic series were “ staging procedures ”: limited sampling without division of lumbar vessels ; the procedure stopped if metastases were found and chemotherapy followed. Later series: lower node yields than open series ; adjuvant chemotherapy rates higher than expected after a proper open RPLND, often given for pathological stage IIA , where observation is preferred	Dual therapy in early-stage disease; chemorefractory viable GCT or teratoma left behind → worse cancer-specific outcomes, including late relapse

Concern	Basis	Consequence
2. Inappropriate patient selection	MIS surgeons without GCT expertise operating on clinical stage IA patients who should be observed , the rare exception being teratoma with secondary somatic transformation in the primary; many stage IB patients; pN0 rates up to 83%	Unindicated surgery
3. Intraoperative safety	Surgery around the great vessels; surgeon absent from the bedside at the console; robot-trained surgeons may lack open skills for emergency conversion. Conversion ≈ 5%	Potentially catastrophic haemorrhage
4. Oncological uncertainty	No large series beyond 5 years; abnormal metastatic patterns (including peritoneal carcinomatosis) reported from GCT centres, attributed to poor technique and pneumoperitoneum	Greater treatment burden, poor cancer outcomes; late relapse unassessable
5. Resource use	Higher cost and longer operative time	—

TABLE 81.1 PERIOPERATIVE AND SHORT-TERM ONCOLOGIC OUTCOMES OF MINIMALLY INVASIVE RPLND (REBUILT: RANGES ACROSS SERIES)

Outcome	Laparoscopic (7 series, 2001–2012)	Robotic (13 series, 2015–2022)
Operative time (min)	≈180–320	≈150–340 (not given for several series)
Conversion to open	0–7%	0–17% (most 0–8%)
Estimated blood loss (mL)	<100–390	50–≈900 (most ≤200)
Length of stay (days)	1–6	1–6 (most 1–3)
Major complication	0–21%	0–35% (most ≤15%)
Follow-up (months)	15–84	4–49
Node yield	≈10–26	7–33 (recent series ≈30)
Adjuvant chemotherapy	Frequent — up to about one-third	0–18%
Recurrence	0–10%	0–11%

- **What the table is for:** the MI-RPLND evidence base — entirely **case series**, no comparator arm.
- **Pattern:** **short stay and modest blood loss** throughout; **node yields rise in the recent robotic series toward the open benchmark (≈30)**; **adjuvant chemotherapy falls** from the laparoscopic to the recent robotic era; **follow-up is short**, and shorter in robotic series.
- **The trap:** low recurrence rates look reassuring, but are **confounded by high pN0 rates and adjuvant chemotherapy** — they do not prove MI-RPLND is oncologically equivalent to open RPLND.

Concept — *Every criticism of MI-RPLND reduces to one fear: that the **approach changes the operation** — a lesser dissection, the wrong patient, or an unsafe setting — and so trades the **single-modality cure** that justifies RPLND for a smaller scar.*

RATIONALE FOR MINIMALLY INVASIVE RETROPERITONEAL LYMPH NODE DISSECTION

- **Reported benefits:** **less blood loss, better convalescence, better cosmesis, quicker return to normal activity**, and **consistently shorter hospital stay** than open RPLND.
- **Theoretical benefit:** **fewer incisional hernias**, extrapolated from other open-versus-MIS comparisons.
- **Counter-argument:** the benefits may be only **incremental**. Without **level 1 evidence including patient-reported outcomes**, it is unknown whether MI-RPLND confers **no benefit, incremental benefit, major benefit or harm**.

SURGICAL TECHNIQUE

Unchanging principles

- Indication decided by a multidisciplinary team.
- Contemporary staging: axial imaging within ≈4 weeks of surgery; serum tumour markers within 1–2 weeks.
- Surgeon expert in retroperitoneal anatomy and common vascular and visceral variants.

Core operative steps — identical whatever the approach

1. Safe abdominal entry → retroperitoneal exposure
 2. → complete removal of the appropriate nodes by split-and-roll
 3. → prospective identification and sparing of the ejaculatory nerves
 4. → division of lumbar vessels → spermatic cord removal → closure.
- None of these steps may be compromised: MI-RPLND must be a complete duplication of open RPLND.

Extent of dissection

- Completeness, template and nerve-sparing are not altered by approach; tumour laterality and intraoperative findings dictate extent.

TEMPLATE SELECTION BY SETTING

Setting	Template
Stage I and small-volume stage II (chemotherapy-naïve)	Extended unilateral template with nerve-sparing — acceptable, though still debated (Fig. 81.4)
Enlarged nodes found intraoperatively	Consider bilateral full template
Postchemotherapy	Bilateral full template with nerve-sparing = standard of care

Boundaries and completeness

- **Boundaries:** renal hila superiorly, ureters laterally, crossing of the ureter over the common iliac vessels inferiorly, plus the spermatic cord.
- Intentional division of lumbar arteries and veins is required to clear all retroaortic and retrocaval tissue.
- Poor-quality RPLND or mass excision gives worse oncological outcomes and survival than complete RPLND.
- GETUG T93BP: good-risk metastatic NSGCT randomised to bleomycin, etoposide and cisplatin (BEP) ×3 versus etoposide and cisplatin (EP) ×4. Relapses (70%) and deaths (50%) were dominated by patients with no surgery or incomplete RPLND. This supports complete RPLND, not split-and-roll or lumbar division specifically.
- Meticulous lymphostasis prevents chylous ascites.

Nerve-sparing

- Performed when feasible without compromising oncological efficacy.
- Sparing the sympathetic chains and postganglionic sympathetic fibres from L1–L4, identified from both chains and preserved to their coalescence in the hypogastric plexus beneath the inferior mesenteric artery (IMA).
- Typically three efferent fibres from each chain contribute to the hypogastric plexus.

Centre and surgeon

- High-volume GCT centres get the indication, timing and sequence of chemotherapy and surgery right; surgical experience improves outcome.

- Anatomical variation and **postchemotherapy desmoplasia** demand **retroperitoneal and vascular skill**. The MI-RPLND surgeon must be **competent in GCT management, open RPLND and MI-RPLND**.

KEY POINTS

- **Completeness, template and nerve-sparing** are not affected by approach (robotic versus open).
- **Tumour laterality and intraoperative findings** dictate extent.
- **MI-RPLND should completely duplicate open RPLND**.

Concept — *Approach is a means of access, never a determinant of the operation; the oncological operation — boundaries, lumbar division, retrovascular clearance, nerve-sparing — is fixed, and anything less is a staging biopsy.*

PERIOPERATIVE PREPARATION

- **Surgical-anaesthetic plan for major complications** agreed in advance.
- **Prior bleomycin** → review **fluid requirements and oxygenation and ventilation**.
- **Group and save with crossmatch; large-bore intravenous access**.
- **Open instrument sets in the theatre** and available for conversion.
- **Vascular sutures prepared: 4-0 or 5-0 Prolene on an RB needle with a LAPRA-TY ≈4 inches from the needle**.
- **HandPort** systems may allow rapid abdominal access and pressure.
- **Experienced bedside assistant** in complex MIS.
- **Continuous insufflation system** helps if prolonged suction is needed for visibility.
- **Second time-out** to confirm all emergency systems are in place.

PATIENT AND TUMOUR FACTORS

- **No absolute contraindications** to MI-RPLND.
- **Relative contraindications** (as for any MIS): **extensive previous abdominal surgery, morbid obesity**.

CANDIDATE SELECTION — MI-RPLND VERSUS OPEN

Acceptable or ideal for MI-RPLND	Favour open RPLND
Chemotherapy-naïve high-risk stage IB NSGCT electing RPLND	Postchemotherapy RPLND in general
Small-volume stage II: 1–2 enlarged nodes in the primary landing zone, none >3 cm	Large residual mass
Postchemotherapy, ideal: solitary metastasis in the primary landing zone	Large change in mass size pre- to postchemotherapy → likely desmoplastic reaction
Postchemotherapy, ideal: small response to chemotherapy (suggests teratoma), limited desmoplasia	Possible great-vessel resection or reconstruction
—	Less suitable: secondary somatic malignancy; retrocrural disease requiring resection

Concept — *Suitability tracks **predicted dissection difficulty**: small, confined, chemotherapy-naïve or teratoma-like disease suits MIS, whereas **desmoplasia, bulk and vascular involvement** — the hallmarks of postchemotherapy surgery — favour open.*

PATIENT COUNSELLING AND PREPARATION

- **Counsel explicitly on:** bleeding, infection, anejaculation, visceral injury, simple or chylous ascites, hernia, and the laparoscopy- and pneumoperitoneum-specific risks.
- **Conversion is not a default complication** — it may be done electively for **safety or oncological reasons**. However, there is a **theoretical excess risk of major bleeding** if rapid conversion and control fail.
- **High-fat diet the day before surgery** may help **identify lymphatics**.
- **Full bowel preparation is generally unnecessary**; **moderate preparation** may limit gaseous distension.
- **ERAS pathway:** preoperative carbohydrate loading, hydration, non-narcotic analgesia.

TIPS, TRICKS AND TOOLS

- **Robotic fourth arm:** lifts kidney and Gerota's fascia for **paracaval and paraaortic** dissection, then retracts **bowel medially** for **interaortocaval** dissection.
- **Lumbar vessels:** Weck clips on the great-vessel side, vessel sealer on the specimen side.
- **Robotic clip appliers** place clips **parallel to the great vessels**.

KEY POINTS

- Discuss the **common and serious adverse events** of RPLND.
- **Bleeding is rare but potentially catastrophic** and may require conversion.
- **Surgical and anaesthetic teams** agree plans for major complications.

INSTRUMENTATION

INSTRUMENTATION — LAPAROSCOPIC VERSUS ROBOTIC RPLND

Function	Laparoscopic	Robotic
Dissection and energy	Monopolar scissors, bipolar grasper	Monopolar scissors, bipolar graspers
Retraction	Bowel grasper; paddle retractor	Bowel grasper or retractor in the fourth arm — standard
Clips	Laparoscopic clip appliers	Robotic clip appliers — for lumbar vessels and large lymphatics hard to reach from the bedside, or when the assistant is suctioning
Sealing	Ultrasonic shears and sealing devices — unreliable for lymphatics	Vessel sealers for small vessels and ostensibly small lymphatics — no robust data
Suction	Suction-irrigation	Robotic suction device useful
Suturing	Laparoscopic needle drivers	Needle drivers available
Bleeding adjuncts	Laparoscopic mini-laparotomy pads	Additional 12-mm assist ports ready

- **Neither energy platform reliably seals lymphatics** — clips remain dependable.

POSITIONING

- **After induction:** vascular access → **orogastric tube** (bowel distension reduces visibility) → **Foley catheter** → **pad all pressure points**, since long operations risk **nerve injury and rhabdomyolysis**.
- Whatever the plan, **be prepared to perform a bilateral full-template RPLND** if the findings demand it.

POSITIONING OPTIONS FOR MI-RPLND

	Modified lateral decubitus ($\approx 45^\circ$)	Supine, maximal table tilt (laparoscopic)	Supine, steep Trendelenburg (robotic)
Use	Laparoscopic or robotic unilateral template (Fig. 81.2)	Laparoscopic, unilateral or bilateral	Robotic, any template (Fig. 81.1)
Ports	As for renal surgery , slightly more caudal (iliac nodes, cord); 1–2 assistant ports (Fig. 81.3)	Four equally spaced 12-mm trocars (10-mm inner diameter), first 1 cm below the xiphoid	Camera 4–6 cm below the umbilicus
Bowel retraction	Gravity plus bed rotation → substantial medial rotation	Gravity , tilting right and left	Cranial displacement of bowel
Benefits	Familiar from transperitoneal renal surgery; easy conversion to supine by rotating fully toward the elevated side (easier than full lateral decubitus)	Bilateral dissection without extra ports or repositioning	Access to the entire retroperitoneum
Drawbacks	Bilateral dissection usually needs full repositioning and new ports (except in slender patients)	Retraction limited by table tilt ; midline ports suboptimal in obesity	Unfamiliar for retroperitoneal surgery

- **Entry: Veress needle or Hasson technique**; ports placed **under direct vision**. Laparoscopic **12-mm trocars** give versatility (sutures, suction–irrigation, camera mobility).

Figure legends

- **Fig. 81.1** — Steep Trendelenburg: **head, knee and arm padding, strap across the chest, friction pad to prevent sliding**.
- **Fig. 81.2** — Flank position: padded beneath; **lower leg bent, upper leg straight**; arms slightly bent to prevent **hyperextension and ulnar nerve injury**.
- **Fig. 81.3** — Left-template ports: assistant port may sit **above the camera port**, or a **second assistant port** may be added.
- **Fig. 81.4** — Modified templates: **para-aortic nodes above the IMA are taken in both templates**.
- **Fig. 81.5** — Split-and-roll: enter the packet **over the aorta** and **roll off the IVC**; **lateral traction** gives optimal posterior dissection.

APPROACH TO RIGHT-SIDED TEMPLATE DISSECTION

Set-up

- **Right side up** (left modified lateral decubitus, or supine with left tilt). **30° lens in the second-most cranial port**; **scissors in the cranial-most port**; **bipolar in the left hand**; **inferior-most port retracts** bowel medially or kidney laterally.
- **Liver retractor** helpful. Retraction: laparoscopic **paddle**; robotic **fourth arm** with an atraumatic retractor (Small Graptor, ProGrasp or Cadiere forceps).

Sequence

1. **Incise the white line of Toldt** from caecum around the hepatic flexure to the **foramen of Winslow**, dividing the root of the small-bowel mesentery → **Kocherise the duodenum**.
2. **Separate ureter from spermatic cord**; **formal ureterolysis** may be needed postchemotherapy. Fourth arm lifts ureter, kidney and Gerota's fascia for paracaval work, then retracts bowel medially for interaortocaval and paraaortic work.

3. **Gonadal vein clipped and divided at its ostium**, traced to the **internal ring**; the distal cord stump is usually removed at the end, after moving ports one slot inferiorly.
4. **Split over the right common iliac artery** at the ureteric crossing, caudal → cranial to the IVC (Fig. 81.5) → continue up the IVC to the **renal hilum**. **Clip retropancreatic and retroduodenal lymphatics at the left renal vein**.
5. **Roll the paracaval packet laterally** → sympathetic chain found **lateral to the lumbar vessels**; preserve L1–L4 postganglionic fibres → **divide lumbar vessels** (Weck clips on the great-vessel side) → free the packet from the **psoas fascia** up to the right renal hilum.
6. **Know right renal artery–vein relationships from imaging**; preserve accessory lower-pole arteries anterior to the IVC.
7. **Extend the split to the left renal vein** → **roll medially off IVC and aorta** (interaortocaval packet) → divide medial lumbar vessels → lift the packet off the **anterior spinous ligament** from the bifurcation to the renal hilum. **Do not ligate the right renal artery** in the interaortocaval space, especially when it lies at the lower edge of the left renal vein.
8. **Paraortic (above the IMA)**: roll laterally → preserve sympathetic fibres → retract ureter laterally → ligate lumbar vessels → free from psoas to the left renal vessels.
9. **Closure**: haemostatic and lymphostatic agents → complete cord dissection → **reduce pneumoperitoneum to check haemostasis and lymphostasis** → return colon → remove ports under vision → **close fascia at the extraction site and any port ≥10 mm** → **no routine drain**.
- **Tumour-spill precautions**: bag each packet (**Endo Catch**) as soon as it is freed; handle suspicious nodes delicately to avoid **rupture or aerosolisation** of tumour cells.

APPROACH TO LEFT-SIDED TEMPLATE DISSECTION

- **Left side up** (right modified lateral decubitus, or supine with right tilt); **no liver retractor** usually.
1. **Medialise the descending colon**, and also the **spleen and tail of pancreas**, for medial rotation of the mesocolon.
 2. **Separate ureter from gonadal vein**; **divide the gonadal vein at the left renal vein**; trace and remove it to the internal ring.
 3. **Start as for a renal operation**: dissect the **hilum as the superior limit**; control the **lumbar vein draining into the left renal vein**; clip the major lymphatics around the **left renal vein**, which is rich in lymphatics from the **superior mesenteric artery trunk**.
 4. **Split from the ureteric crossing of the left common iliac artery** cranially along the aorta to the left renal vein → **roll laterally** (paraortic and common iliac packets) → divide lateral lumbar vessels → preserve sympathetic fibres → free from **psoas fascia and diaphragmatic crus**.
 5. **Rotate the packet medially** into the interaortocaval space → divide medial lumbar vessels (for a template, **only above the IMA**); the **IMA may be ligated** to aid mesocolic rotation → roll off the **anterior spinous ligament**.
 6. If the template extends, handle **precaval, paracaval and retrocaval** nodes as on the right.

RIGHT VERSUS LEFT TEMPLATE — THE CONTRASTS

	Right	Left
Mobilisation	Toldt from caecum → hepatic flexure → foramen of Winslow; Kocherise duodenum	Descending colon plus spleen and pancreatic tail
Liver retractor	Helpful	Not usually needed

	Right	Left
Gonadal vein	Divided at its ostium (IVC)	Divided at the left renal vein
Split begins	Over the right common iliac artery → IVC	Over the left common iliac artery → aorta
Lymphatic hotspot	Retropancreatic and retroduodenal tissue at the left renal vein	Left renal vein (SMA-trunk lymphatics); lumbar vein into the left renal vein
Named vascular hazard	Right renal artery in the interaortocaval space; accessory lower-pole arteries	Lumbar vein into the left renal vein; IMA may be ligated

BILATERAL TEMPLATES AND STEEP TRENDLENBURG APPROACH

- **Flank or tilt positions:** complete the ipsilateral side → **tilt the other way or reposition** for the contralateral side.
 - **Steep Trendelenburg** is newer: unlike lateral approaches (extensions of laparoscopy), its **novel positioning facilitates bilateral dissection**.
1. **Supine, arms padded and tucked** → **steep Trendelenburg** as for deep pelvic surgery, so bowel falls cranially.
 2. Pneumoperitoneum → **camera 4–6 cm below the umbilicus** → **two robotic arms on the patient's left, one on the right**; scissors right hand, bipolar left, **atraumatic bowel grasper in the fourth arm**; assist port right or left by laterality.
 3. **0° lens first, then 30° down. Docking: da Vinci Si over the left shoulder; Xi from the side.**
 4. **Incise the posterior peritoneum from the ligament of Treitz around the caecum.**
 5. **Peritoneal hammock:** cut peritoneal edges fixed to the anterior abdominal wall (**straight needles or running self-locking sutures**), holding up small bowel, duodenum and both colons.
 6. **Right, left or bilateral template without repositioning.**

Concept — *Positioning is a **trade between familiarity and reach**: flank positions borrow renal-surgery habits but confine the surgeon to one side, whereas steep Trendelenburg with a peritoneal hammock opens the whole retroperitoneum at the cost of an unfamiliar view.*

ADVERSE EVENT MITIGATION AND MANAGEMENT

- **Most feared: major vascular injury.** Prevention: surgeon proficient in **GCT management, open RPLND and MIS**; vascular sutures, blood products and open sets confirmed; **qualified bedside assistant**.

Major bleeding — sequence

1. **Temporise** with pressure (robotic instrument or assist port) plus **increased pneumoperitoneum**.
2. → **Prepare for conversion**; notify anaesthesia; blood products; **consultant urology or vascular help** as needed.
3. → If not controllable by MIS: **keep holding pressure, undock, convert to open**.

If managed minimally invasively

- **Blood on the camera** → **single-eye setting** on the robot, or **move the camera to another port**.
- **Extra trocar** for exposure, pressure and suction plus a second assist instrument; laparoscopic sponges and laparotomy pads.
- **Identify, understand and isolate the injury** (avulsed lumbar, cavotomy, aortotomy). **No blind suturing or clipping**. Vascular sutures with **LAPRA-TY** speed repair.

Lymphostasis

- **Lower pneumoperitoneum at the end** to reveal leaks → suture or clip. This **may still miss leaks**, because the lymphatic system is **low-pressure**.
- **Liberal clips** around lymphatic-dense sites: **left renal vein** and the **inferior mesenteric vein crossing anterior to it**.
- **Bipolar and electrosealing devices** may help (extrapolated from pelvic lymphadenectomy); **sealants** (Tisseel, Vista Seal) at completion.

Concept — *Crisis management follows **temporise** → **prepare** → **decide**: pressure buys time, preparation makes conversion controlled rather than desperate, and **no manoeuvre is performed on an injury not yet seen**.*

POSTOPERATIVE MANAGEMENT

- **Extubate and remove the gastric tube**; in shorter cases **remove the catheter before recovery**.
- **Early ambulation, early diet, multimodal non-narcotic analgesia, ERAS**.
- **Discharge typically within 1–2 days**.

DATA AND OUTCOMES FOR MINIMALLY INVASIVE RETROPERITONEAL LYMPH NODE DISSECTION

- Evidence is limited to **single- and multi-institutional case series** (Table 81.1).

Safety

- **Complication profile broadly consistent with open surgery**.
- **Brisk bleeding** is the complication unique to MI-RPLND → conversion; **conversion ≈ 5%**.
- **Antegrade ejaculation preserved in ≥90%** after primary MI-RPLND.
- **Chylous ascites rates commensurate with open RPLND**.
- **Postchemotherapy safety data: small series only — no meaningful conclusions**.

Efficacy

- **Chemotherapy-naïve RPLND must be diagnostic and therapeutic**.
- **Benchmark: recurrence-free survival ≈85%** for limited pathological node-positive disease **without adjuvant chemotherapy**. Large contemporary high-volume series show **≈80% regardless of pN1, pN2 or pN3**, which questions **reflex adjuvant chemotherapy for pN2–pN3**.
- **Recurring criticisms: (1) unindicated surgery** in stage IA (arguably IB); (2) **diagnostic-only intent with low node yields**; (3) **very high adjuvant chemotherapy use**.
- **Consistent themes across series: follow-up <5 years**; node yields variable but **approaching the open benchmark of ≈30**; **node-positive rates lower than expected** for stage IB–II; **adjuvant chemotherapy too frequent**; **recurrence acceptable, probably because of high pN0 rates and adjuvant chemotherapy**.
- **Postchemotherapy efficacy data: small series only — no meaningful conclusions**.

KEY POINTS

- **No randomised data** compare perioperative outcomes or long-term efficacy of open versus robotic RPLND.
- **Node yields in recent series approach the open benchmark**, though historical MI series were lower.

- **RPLND should always be performed with therapeutic intent.**

Concept — *MI-RPLND's outcome data are **reassuring but uninterpretable**: acceptable recurrence is bought partly by operating on node-negative patients and adding chemotherapy — the very practices that defeat the purpose of primary RPLND.*

FUTURE DIRECTIONS

- RPLND remains a **mainstay in both chemotherapy-naïve and postchemotherapy settings**, and indications are **expanding** as patients and clinicians seek to **avoid systemic chemotherapy and abdominal radiotherapy** and their long-term harms.
- **Stage II seminoma**: renewed interest in RPLND, with **excellent complication profiles and acceptable oncological outcomes**.
- **Circulating microRNA miR-371a-3p** — a **sensitive and specific biomarker** that may:
 - identify **occult metastases in stage I** patients who would benefit from RPLND;
 - identify **clinically small-volume nodes that are pN0**, sparing RPLND;
 - select patients for **adjuvant chemotherapy after RPLND**.
- **A randomised open-versus-robotic trial is unlikely** (recruitment, community support) → rely on **well-conducted prospective registries**.
- **Three conditions for MI-RPLND**: (1) **multidisciplinary decision**; (2) **MIS duplicates open surgery** in extent, nerve-sparing and indication; (3) surgeon expert in **GCT management, open RPLND and MIS**.

Concept — *The future of RPLND lies in **better selection rather than better access**: biomarkers such as miR-371a-3p promise to operate only on those who harbour disease, which is precisely the weakness MI-RPLND series have exposed.*